

A Hybrid Framework for AI-Generated Media Authentication: Fusing Content Provenance, Invisible Watermarking, and Deep Learning Forensics

Introduction to the Evolving Threat Landscape

The rapid proliferation and refinement of generative artificial intelligence have fundamentally altered the digital media landscape. Models based on latent diffusion architectures, generative adversarial networks (GANs), and large language models (LLMs) are now capable of synthesizing photorealistic images, highly convincing audio, and temporally coherent video at scale. While these advancements drive innovation across creative and industrial sectors, they simultaneously present severe challenges to information integrity, identity verification, and digital forensics. The boundary between authentic and synthetic media has become imperceptible to the human visual and auditory systems, necessitating the deployment of robust, automated detection and authentication frameworks¹.

Historically, the detection of manipulated media—colloquially termed deepfakes—has relied on reactive, single-layer approaches. These have primarily consisted of deep neural networks trained to identify spatial artifacts, spectral anomalies, or temporal inconsistencies in manipulated media³. However, the continuous evolution of generative algorithms creates a structural generalization gap. Detectors trained on contemporary datasets rapidly degrade when exposed to media from unseen, newer-generation models, a phenomenon exacerbated by adversarial laundering techniques such as format conversion, noise injection, and diffusion-based purification⁵. Furthermore, active detection models alone cannot establish the authoritative origin of a piece of media; they can only assign a probabilistic score indicating the likelihood of manipulation based on statistical deviations from a training distribution.

To address these compounding vulnerabilities, the authentication paradigm must shift from reactive detection to proactive, multi-layered provenance. This report proposes and extensively details a hybrid framework for authenticating AI-generated media, directly addressing the limitations of isolated detection systems⁷. By synthesizing cryptographically verifiable content provenance, invisible digital watermarking, and advanced deep-learning-based forensic analysis, the framework establishes a defense-in-depth architecture. Because each of these signals operates under fundamentally different threat models and degradation curves, their integration requires sophisticated evidence fusion. Utilizing Dempster-Shafer Theory (DST), the proposed system mathematically combines independent, and occasionally conflicting, authentication signals into a unified verification status, overcoming the structural limitations inherent in any single approach⁸.

Regulatory Directives and the Content Provenance

Protocol

The imperative for verifiable media authentication is no longer purely an academic or corporate security concern; it is increasingly driven by sweeping regulatory mandates designed to preserve epistemic trust in digital ecosystems. Most notably, Article 50 of the European Union Artificial Intelligence Act (EU AI Act) enforces stringent transparency obligations on the providers and deployers of generative AI systems. Effective August 2, 2026, the legislation mandates that outputs from generative systems—whether audio, image, video, or text—must be marked in a machine-readable format and detectable as artificially generated or manipulated¹⁰. The marking mechanisms deployed to satisfy this regulation must meet specific statutory criteria: they must be effective, interoperable, robust, and reliable as far as is technically feasible¹². Failure to comply exposes organizations to maximum penalties of up to €15 million or 3% of total worldwide annual turnover¹¹.

The Coalition for Content Provenance and Authenticity (C2PA)

To meet these complex regulatory requirements, the technology and media industries have standardized around the Coalition for Content Provenance and Authenticity (C2PA) specification. C2PA establishes an open, cryptographic framework for recording the provenance of digital media from the exact point of capture or generation through all subsequent editorial edits, algorithmic transformations, and platform distributions¹⁴. The architecture is built upon the embedding of a tamper-evident record, termed a manifest, directly into the media file's metadata container.

The C2PA manifest operates through three primary, cryptographically linked components: assertions, claims, and claim signatures¹⁶. Assertions are specific, granular statements about the content. These can encompass metadata such as the software used for creation (e.g., utilizing the `c2pa.created` action label), specific editing actions like cropping or color grading (`c2pa.actions.v2`), or the explicit declaration of AI generation¹⁸. These individual assertions are serialized into Concise Binary Object Representation (CBOR) format for extreme data compactness¹⁶. The serialized assertions are then hashed and packaged into a comprehensive data structure known as a claim. Finally, this claim is sealed with a digital signature utilizing Public Key Infrastructure (PKI). The signature adheres to the CBOR Object Signing and Encryption (COSE) standard (RFC 9052), which utilizes X.509 certificate chains to ensure that the claim and its underlying assertions are genuinely from the stated source and have not been altered in transit¹⁶.

The physical embedding of this cryptographic data relies on the JPEG Universal Metadata Box Format (JUMBF), a component of the JPEG Systems standards that provides a structured way to manage metadata within binary files²⁰. The JUMBF architecture provides a hierarchical container system comprising a Super Box, which acts as the primary envelope; a Description Box, which details the metadata type, format, and purpose via Universally Unique Identifiers (UUIDs); and a Data Box, which houses the actual serialized CBOR assertions and COSE signatures²⁰. This standardized structure allows non-compliant software, such as legacy image viewers, to simply ignore the provenance data without error, while ensuring that modern, compliant validators can parse the byte layout and authenticate the historical chain of

custody¹⁵.

Hard Bindings, Soft Bindings, and the Metadata Stripping Challenge

A critical vulnerability in any metadata-based provenance architecture is the potential detachment of the manifest from the underlying media asset, whether accidental or adversarial. To counter this, the C2PA specification mandates the inclusion of a "hard binding" within the claim. A hard binding is a cryptographic hash, typically employing the SHA-256 algorithm, computed over a specific, defined byte range of the media file¹⁶. Crucially, this hashing operation explicitly excludes the JUMBF region containing the manifest itself to prevent recursive hashing errors. When a verifier inspects the file, it recomputes the hash over the visual or audio payload. If a single pixel or audio sample has been altered outside of a C2PA-compliant editing tool, the newly computed hash will fail to match the stored hard binding hash, rendering the manifest invalid and making the tampering immediately cryptographically evident²².

However, the modern digital ecosystem is fundamentally hostile to embedded metadata. Major social media networks and content delivery networks automatically strip EXIF, XMP, and JUMBF metadata during their upload and transcoding pipelines²³. This stripping is performed to reduce payload sizes for bandwidth optimization and to scrub potentially sensitive user information, such as geolocation coordinates. When the JUMBF container is discarded by these platform pipelines, the hard binding is irrevocably lost, and the C2PA manifest is entirely unrecoverable from the file header alone. This reality introduces a severe structural limitation for content authenticity on the open web.

To ensure the durability of provenance data across hostile distribution channels, the C2PA specification introduces the concept of "soft bindings." A soft binding functions as a decoupled recovery index, bypassing the fragile file header entirely by embedding an identifier directly into the perceptual data of the media—specifically, the pixel values of an image or the waveform of an audio file²⁴. When the file header is stripped, a validator can extract the soft binding from the media's raw content and query a remote, cloud-based C2PA Manifest Store to retrieve the original, detached manifest²³. This symbiotic relationship fundamentally necessitates the integration of highly robust invisible digital watermarking into the overarching provenance architecture, bridging the gap between fragile cryptographic headers and resilient visual payloads.

Invisible Digital Watermarking in Generative Models

While C2PA provides the cryptographic structure for establishing trust, invisible watermarking anchors that trust directly into the media content itself. Generative watermarking diverges significantly from traditional post-hoc steganography. Rather than hiding a message in pre-existing media using simple spatial domain alterations, modern generative watermarking often integrates the signal directly into the generative process of the neural network, manipulating the latent space or the probability distributions to embed information natively²⁶.

Latent Space Embedding and Gaussian Shading

In diffusion models, embedding watermarks directly into the latent space during the denoising process ensures that the signal is inherently tied to the structural semantics of the generated

image, rather than merely layered on top of the pixel data. A prominent and mathematically elegant technique in this domain is Gaussian Shading, a training-free, performance-lossless watermarking mechanism designed specifically for latent diffusion models (LDMs)²⁸.

LDMs, such as the widely deployed Stable Diffusion architectures, synthesize images by progressively removing Gaussian noise from a lower-dimensional latent representation over a series of timesteps. This denoised latent representation is eventually mapped back to high-resolution pixel space via a Variational Autoencoder (VAE) decoder²⁶. Gaussian Shading embeds the watermark by mapping the signal to latent representations that strictly follow a standard Gaussian distribution. Because the watermarked latent variables are statistically indistinguishable from the non-watermarked latent variables drawn from the model's natural prior, the generative performance of the model remains completely unaffected, preserving the exact Fréchet Inception Distance (FID) and Image Reward scores of the base model²⁹.

The mathematical formulation relies on spreading the watermark information throughout the latent representation via predefined channel diffusion factors and height-width diffusion factors. During the generation process, the watermark information diffuses into the global semantics of the image, achieving exceptional robustness against localized cropping and filtering²⁹. In the extraction phase, the target image undergoes Denoising Diffusion Implicit Model (DDIM) inversion. This process implicitly predicts the noise at each reverse time step, reconstructing the deterministic noise trajectory to allow the latent vector to be accurately retrieved from the pixel space²⁹. The watermark is then extracted by applying a Discrete Cosine Transform (DCT) to the designated high-frequency channels of the retrieved latent vector. The original payload is recovered by calculating the correlation between a known spread-spectrum code and the extracted detection matrix³⁰. Harnessing the extensive scope of the Stable Diffusion latent space, Gaussian Shading achieves a high-capacity watermark of 256 bits, vastly outperforming prior methods that were limited to 32 or 64 bits²⁹.

Neural Network Watermarking Protocols: InvisMark

For high-resolution generative outputs where extensive payload capacity and extreme imperceptibility are non-negotiable requirements, specialized neural encoder-decoder architectures operate as highly optimized post-generation embedding networks. A state-of-the-art implementation of this approach is InvisMark¹.

InvisMark utilizes a Multi-modal Unsupervised Image-to-image Translation (MUNIT)-based architecture for its encoder. This design incorporates complex skip connections, ensuring that fine-grained visual details are perfectly preserved during the embedding of the binary tensor payload into the cover image. The decoder module, responsible for extracting the payload, abandons traditional ResNet architectures in favor of the ConvNeXT framework. ConvNeXT was selected due to its superior training stability and enhanced feature extraction robustness when subjected to heavy image distortion³³. A defining characteristic of InvisMark is its capacity to reliably encode massive 256-bit payloads while preserving image quality. This extended capacity is critical, as it facilitates the embedding of complex identifiers, such as standard 128-bit Universally Unique Identifiers (UUIDs), paired directly with highly redundant Bose-Chaudhuri-Hocquenghem (BCH) error correction codes³³.

To achieve state-of-the-art imperceptibility—characterized by a Peak Signal-to-Noise Ratio (PSNR) approaching 51 and a Structural Similarity Index Measure (SSIM) of approximately 0.998—alongside a bit recovery rate exceeding 97% under severe distortion, the network is trained using a paradigm known as Compositional Adversarial Training (CAT)¹. Standard robust watermarking typically trains against random noise augmentations, which frequently fail to represent the combinatorial space of realistic attack pipelines³⁴. Conversely, the CAT protocol subjects the watermark to autoregressive sequences of worst-case adversarial transformations during the training loop. The optimization process utilizes a highly tuned weighted loss function that strictly balances pixel-level error in the YUV color space, Learned Perceptual Image Patch Similarity (LPIPS) to mimic human visual perception, and the cross-entropy loss of the watermark recovery accuracy³³.

Logit Modulation and Statistical Thresholding: SynthID

For text and continuous stream generation, where the output consists of discrete tokens rather than continuous pixel values, watermarking requires a fundamentally different theoretical approach. Google DeepMind’s SynthID represents the current industrial standard for generation-time statistical watermarking in Large Language Models³⁵.

In standard text generation, LLMs sample the next token from a probability distribution (logits) computed over the model's entire vocabulary. SynthID intervenes directly in this final sampling stage via an algorithmic mechanism known as Tournament Sampling. A pseudorandom ^g-function, seeded jointly by a secret cryptographic key and the specific context of the preceding tokens, modulates the likelihood scores of the candidate tokens proposed by the LLM³⁵. The candidates then compete in a multi-layer tournament where the pseudorandom scoring functions systematically advance tokens that statistically correlate with the secret watermark key³⁵.

Because the ultimately selected token is still drawn from the subset of highly plausible semantic candidates proposed by the underlying model, the semantic quality, perplexity, and fluency of the text are preserved, rendering the technique non-distortionary³⁵. Detection relies on computing a Bayesian or Weighted Mean score across the sequence of generated tokens without requiring access to the original LLM's weights. If the aggregate score exceeds a statistically calibrated threshold, designed to strictly control the false positive rate, the text is positively flagged as watermarked³⁵. While highly scalable and currently deployed in consumer applications like Google Gemini, statistical text watermarking remains inherently vulnerable to specific semantic attacks. Malicious actors utilize paraphrasing attacks via secondary LLMs, translation attacks, and emoji-insertion attacks to disrupt the contiguous sequence of tokens, thereby lowering the statistical correlation below the detection threshold³⁸.

To summarize the capabilities of the primary generative watermarking technologies, the following comparison highlights their distinct operational domains and capacities.

Technology Framework	Operational Domain	Embedding Mechanism	Primary Advantage	Payload Capacity
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Gaussian Shading	Latent Diffusion (Images)	Latent space standard normal distribution mapping	Provably performance-lossless on generative quality	256 bits
InvisMark	Post-generation (Images)	MUNIT Encoder / ConvNeXT Decoder with CAT	Extreme robustness to compositional transformations	256 bits + BCH
SynthID	LLMs (Text), Audio, Video	Tournament sampling / Logit modulation	Generative quality preservation without structural metadata	Statistical threshold

Adversarial Evasion and the Purification Vulnerability

The persistence of invisible watermarks is continually challenged by the rapid advancement of adversarial removal techniques. While generative watermarks are heavily optimized to survive standard lossy compression (e.g., JPEG, MP3), spatial cropping, and linear filtering, they are highly vulnerable to advanced generative regeneration and diffusion purification attacks⁴¹. Techniques such as DiffPure utilize the exact mathematical mechanisms of generative diffusion models to cleanse images of both adversarial perturbations and embedded watermarking signals⁵. The attack methodology is exceptionally powerful because it requires no knowledge of the specific watermarking algorithm. Given a watermarked image, the attacker passes the media through the forward stochastic differential equation (SDE) of a pre-trained diffusion model. The algorithm adds calculated Gaussian noise up to a specific, shallow diffusion

timestep (e.g., $t^* = 0.3$). This controlled noise addition acts as a low-pass filter in the latent space, effectively destroying the high-frequency and tightly coupled structural patterns of the embedded watermark⁵.

Once the watermark signal is sufficiently degraded, the image is reconstructed by solving the reverse generative SDE using an adjoint method for computational efficiency. The reverse process synthesizes a "clean" image that retains the original semantic and visual content to a high degree of fidelity, but entirely lacks the statistical watermarking signal⁵. Furthermore, recent analyses published in major computer vision conferences demonstrate that if an adversary has access to a quality oracle and a perturbation oracle, they can actively erode the

watermark boundary using stochastic erosion or latent space inversion, effectively restoring total deniability to the deepfake⁴⁴.

This vulnerability mathematically proves a critical point for system design: invisible watermarking, while essential for soft-binding C2PA manifests, cannot be relied upon as a singular point of failure. When an adversary actively scrubs the C2PA JUMBF container via platform upload, and subsequently sanitizes the pixel payload via a diffusion purification attack, both the cryptographic and steganographic layers of the authentication framework are neutralized. In this scenario, the system must cascade to the final layer of defense: blind, deep learning-based forensic detection.

Deep Learning-Based Forensic Detection

When cryptographic provenance is stripped and watermarks are adversarially purified, the authentication framework must rely entirely on deep learning forensics. This layer analyzes the intrinsic, unintentional artifacts left behind by the generative models themselves.

Spatial Artifact Analysis via Convolutional Neural Networks

Early generations of deepfakes, particularly face-swapping algorithms utilizing autoencoders and early GAN architectures, introduced prominent spatial inconsistencies. These included rigid blending boundaries, unnatural eye reflections, mismatched skin tones, and severe upsampling artifacts resulting from resolution mismatches between the generated face and the background frame³. Convolutional Neural Networks (CNNs), particularly the XceptionNet architecture, have proven highly effective at localizing these spatial anomalies³.

XceptionNet modifies standard inception modules by employing depthwise separable convolutions. This approach mathematically decouples spatial correlations from cross-channel correlations, which drastically reduces the parameter count of the network while significantly enhancing its feature extraction capabilities for subtle high-frequency noise patterns⁴⁶. When trained on extensive benchmark datasets, XceptionNet achieves exceptional classification metrics. Through the utilization of Gradient-weighted Class Activation Mapping (Grad-CAM) and Shapley Additive Explanations (SHAP), the network can visualize the exact spatial regions—such as irregular skin textures or blurred jawline boundaries—that trigger the manipulation classification, providing essential explainability to the forensic output⁴⁸.

To demonstrate the baseline efficacy of spatial CNNs, the following table summarizes the performance of various architectures across standard deepfake detection benchmarks.

Detection Architecture	FaceForensics++ (AUC)	Celeb-DF V2 (AUC)	WildDeepfake (AUC)
XceptionNet (Base)	98.00%	89.91%	~65.00%
EfficientNet B4	99.10%	91.50%	~68.00%

DFDT (Vision Transformer)	99.41%	99.31%	81.35%
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Despite high performance on controlled datasets like FaceForensics++, spatial-only CNNs suffer from severe performance degradation when confronted with heavily compressed media or media generated by entirely unseen architectures, highlighting the necessity for temporal analysis⁶.

Spatiotemporal Analysis in Video

Video deepfakes possess a critical temporal dimension that single-frame spatial detectors completely ignore. Even if an individual frame is flawlessly generated by a modern diffusion model, maintaining absolute temporal coherence—ensuring consistent micro-expressions, accurate physiological blinking rates, and stable background illumination across dozens of consecutive frames—remains a profound challenge for current generative technologies⁴. To exploit these temporal artifacts, the authentication framework utilizes 3D Convolutional Neural Networks (such as R3D-18) and hybrid architectures that combine spatial CNN extractors with sequence modeling networks⁴. In a highly optimized hybrid pipeline, a lightweight CNN, such as MobileNetV2 pre-trained on ImageNet, extracts high-level spatial embeddings from a sequence of frames, providing an optimal trade-off between inference speed and representation capacity⁵¹. These embeddings are simultaneously passed into a Bidirectional Long Short-Term Memory (BiLSTM) network to capture short-term temporal incongruities—such as jitter or inconsistent mouth movements—and a Transformer encoder equipped with self-attention mechanisms to model long-term global temporal relationships across the entire video clip⁵¹.

By optimizing the network using a composite loss function that combines standard binary cross-entropy with a stringent temporal-consistency regularizer, the model learns to heavily penalize unnatural phase shifts in the temporal spectrum⁴. Empirical evaluations demonstrate that these temporal artifacts survive social media re-encoding much better than high-frequency spatial artifacts. For example, high-quality GAN output can reduce spatial-only detection accuracy by several percentage points, while leaving temporal inconsistencies largely intact, providing a highly robust detection signal for video forensics in hostile environments⁴.

Zero-Shot Generalization and Transferability

The most critical failure mode of deep learning forensics is the cross-generator generalization gap. Detectors that achieve near-perfect Area Under the Curve (AUC) scores on academic benchmarks frequently collapse to random chance when deployed in the wild against media generated by novel, unseen diffusion models. Recent in-the-wild evaluations report AUC drops of 45–50% for state-of-the-art open-source models⁶. These models tend to overfit to generator-specific high-frequency artifacts—such as the specific checkerboard patterns produced by a transposed convolution layer in a known GAN—rather than learning fundamental semantic indicators of authenticity⁵³.

To achieve true zero-shot generalization, recent advances leverage frozen visual foundation models, most notably DINOv3⁵³. DINOv3, trained via image-only self-distillation rather than text-image pairing, intrinsically learns to represent globally coherent, low-frequency structural representations that are invariant to localized noise and specific architectural artifacts. Research indicates that utilizing a lightweight linear probe on top of frozen DINOv3 token representations captures highly transferable authenticity cues, achieving 87.5% accuracy on out-of-distribution deepfake datasets using minimal training data⁵³.

Furthermore, techniques like FakeInversion actively invert suspicious images back into the latent space of an open-source Stable Diffusion model. By tracking the trajectory of the feature maps during DDIM inversion, the model calculates the divergence between the image's inherent noise structure and the expected natural image distribution. This creates a powerful, generalized synthetic image detector that relies on the fundamental physics of the diffusion process, rather than requiring prior exposure to the specific generator used by the attacker⁵⁴.

Multimodal Evidence Fusion using Dempster-Shafer Theory

The isolated limitations of the three authentication pillars dictate that true media verification requires sophisticated evidence fusion. C2PA provides definitive proof but is susceptible to metadata stripping; invisible watermarking provides resilient soft bindings but is vulnerable to diffusion purification; deep learning provides an unremovable forensic net but outputs probabilistic scores susceptible to false positives. To fuse binary cryptographic signals with probabilistic neural network outputs, the framework requires a mathematical system capable of handling uncertainty, absolute ignorance, and directly conflicting evidence.

Dempster-Shafer Theory (DST) is uniquely suited for this task, as it allows degrees of belief to be assigned to sets of propositions rather than forcing a strict probability distribution over mutually exclusive singletons, avoiding the rigid prior assumptions required by traditional Bayesian inference⁸.

Mathematical Formulation of DST

In the context of this authentication framework, the Frame of Discernment (Ω) represents the mutually exclusive states of the media's origin. We define the frame as

$$\Omega = \{\text{Authentic}, \text{Synthetic}\}$$

The power set of Ω , denoted as 2^Ω , encompasses all possible combinations of these states: $\{\emptyset, \{\text{Authentic}\}, \{\text{Synthetic}\}, \{\text{Authentic}, \text{Synthetic}\}\}$.

A Basic Probability Assignment (BPA), or mass function m , maps each element of the power set to a value in the interval $[0, 1]$, subject to the absolute constraints that the mass of the empty set is zero ($m(\emptyset) = 0$) and the sum of all masses across the power set exactly equals 1⁸:

$$\sum_{A \subseteq \Omega} m(A) = 1$$

Crucially, the state $\{\text{Authentic}, \text{Synthetic}\}$ (often denoted simply as Ω in the context of the power set) represents *ignorance* or uncertainty. Mass assigned to this state implies that the evidence cannot distinguish between real or fake, a critical feature for handling stripped metadata⁸.

From the mass function, two bounding metrics are derived for any hypothesis A :

1. **Belief** (Bel): The lower bound of certainty, representing the total mass of evidence that strictly and directly supports hypothesis A and its subsets⁵⁷.

$$Bel(A) = \sum_{B \subseteq A} m(B)$$

2. **Plausibility** (Pl): The upper bound of certainty, representing the extent to which the total available evidence fails to refute hypothesis A ⁵⁷.

$$Pl(A) = \sum_{A \cap B \neq \emptyset} m(B)$$

The numerical gap between Belief and Plausibility ($Pl(A) - Bel(A)$) rigorously quantifies the system's uncertainty regarding the specific hypothesis⁸.

Defining Mass Functions for the Authentication Signals

The framework translates the outputs of the three extraction phases into distinct mass functions (m_{prov} , m_{wm} , m_{dl}).

1. Provenance Mass Function (m_{prov}): If a C2PA manifest is present in the JUMBF container, cryptographically valid against the trust list, and contains an assertion of AI generation, the mass function heavily favors the synthetic hypothesis. However, if no manifest is present—the most common scenario on social media—the system does not assume the image is authentic. Instead, it assigns the entirety of the mass to the uncertainty state, perfectly modeling the reality of metadata stripping²³.

- *Valid AI Manifest:* $m_{prov}(\{\text{Synthetic}\}) = 0.95$, $m_{prov}(\Omega) = 0.05$
- *Stripped/No Manifest:* $m_{prov}(\Omega) = 1.0$

2. Watermark Mass Function (m_{wm}): Invisible watermarks, such as Gaussian Shading or SynthID, are highly indicative of synthetic generation if successfully extracted and validated. However, their absence does not guarantee authenticity, as they may have been erased via DiffPure. If a watermark is detected with high bit accuracy, a strong mass is assigned to the synthetic state. If absent, a massive allowance for ignorance is maintained.

- *Watermark Detected:* $m_{wm}(\{\text{Synthetic}\}) = 0.90$, $m_{wm}(\Omega) = 0.10$
- *No Watermark Detected:* $m_{wm}(\{\text{Authentic}\}) = 0.10$, $m_{wm}(\Omega) = 0.90$

3. Deep Learning Forensics Mass Function (m_{dl}): The ensemble deep learning model (combining spatial XceptionNet and temporal 3D CNNs) outputs a continuous confidence

score p for the media being synthetic. This score is calibrated and mapped directly to the mass function, with a dynamically calculated uncertainty margin based on the statistical variance between the spatial and temporal sub-models⁵⁷.

- Assume the model is 80% confident the media is fake, with a calculated 10% uncertainty margin due to temporal jitter.
- $m_{dl}(\{\text{Synthetic}\}) = 0.80$, $m_{dl}(\{\text{Authentic}\}) = 0.10$,
 $m_{dl}(\Omega) = 0.10$.

Fusing the Evidence via Dempster's Rule

To mathematically combine these independent pieces of evidence, the framework utilizes

Dempster's Rule of Combination⁸. For two independent mass functions m_1 and m_2 , the

combined mass $m_{1\oplus 2}(A)$ for a hypothesis A (where $A \neq \emptyset$) is given by the orthogonal sum:

$$m_{1\oplus 2}(A) = \frac{\sum_{B \cap C = A} m_1(B)m_2(C)}{1 - K}$$

Where K represents the total mass assigned to strictly conflicting evidence (i.e., the intersection of the sets is empty, such as Authentic intersecting with Synthetic):

$$K = \sum_{B \cap C = \emptyset} m_1(B)m_2(C)$$

Applied Case Study: Evasion Attack Resolution Consider a sophisticated, real-world evasion attack: An adversary generates a highly realistic deepfake video. To evade detection, they upload the video to a social media platform, which automatically strips the C2PA JUMBF container²³. Prior to upload, they apply a DiffPure algorithm to aggressively degrade the invisible watermark⁵. The system must now resolve the following extracted mass functions:

1. m_{prov} detects no manifest: $m_{prov}(\Omega) = 1.0$
2. m_{wm} fails to detect the degraded watermark conclusively, assigning a weak signal toward authenticity: $m_{wm}(\{\text{Authentic}\}) = 0.15$, $m_{wm}(\Omega) = 0.85$.
3. m_{dl} (3D CNN temporal analysis) bypasses the spatial purification artifacts and detects heavy temporal inconsistencies in the video sequence: $m_{dl}(\{\text{Synthetic}\}) = 0.85$, $m_{dl}(\{\text{Authentic}\}) = 0.05$, $m_{dl}(\Omega) = 0.10$.

The system sequentially combines the evidence. First, combining m_{prov} and m_{wm} : Because m_{prov} is entirely ignorant (1.0 assigned to Ω), the mathematical combination $m_{prov \oplus wm}$ simply equals m_{wm} .

Next, the system combines the result (m_{wm}) with the forensic signal (m_{dl}). We calculate the intersections of all power set elements:

- $\{\text{Authentic}\} \cap \{\text{Authentic}\} = \{\text{Authentic}\}$: $0.15 \times 0.05 = 0.0075$
- $\{\text{Authentic}\} \cap \Omega = \{\text{Authentic}\}$: $0.15 \times 0.10 = 0.015$
- $\Omega \cap \{\text{Authentic}\} = \{\text{Authentic}\}$: $0.85 \times 0.05 = 0.0425$
 - **Total unnormalized mass for Authentic:** 0.065
- $\Omega \cap \{\text{Synthetic}\} = \{\text{Synthetic}\}$: $0.85 \times 0.85 = 0.7225$
 - **Total unnormalized mass for Synthetic:** 0.7225
- $\Omega \cap \Omega = \Omega$: $0.85 \times 0.10 = 0.085$
 - **Total unnormalized mass for Uncertainty:** 0.085
- **Conflict (K):** The mass assigned to the impossible intersection

$$\{\text{Authentic}\} \cap \{\text{Synthetic}\} = \emptyset \quad ; \quad 0.15 \times 0.85 = 0.1275$$

Normalizing the results by dividing by $1 - K$ ($1 - 0.1275 = 0.8725$):

- $m_{final}(\{\text{Authentic}\}) = 0.065/0.8725 = \mathbf{0.0745}$
- $m_{final}(\{\text{Synthetic}\}) = 0.7225/0.8725 = \mathbf{0.8281}$
- $m_{final}(\Omega) = 0.085/0.8725 = \mathbf{0.0974}$

Through DST fusion, the system logically resolves the conflicting evidence—the absence of a watermark weakly suggesting authenticity versus the deep learning model strongly suggesting forgery. The final system state concludes with an 82.8% belief that the media is synthetic, effectively neutralizing the adversary's multi-layered evasion attack.

Output Classification Matrix

Based on the final fused mass functions and the origin of the driving signals, the system maps the mathematical output to one of four definitive authentication statuses, providing actionable intelligence to the end-user⁷.

Output Status	Mathematical Condition (Post-Fusion)	System Interpretation
Verified AI Origin	$m(\text{Synthetic}) > 0.95$ AND derived from S_{prov} or S_{wm}	Cryptographic proof or intact watermark conclusively confirms generative origin.
Provenance Available	$m(\text{Authentic}) > 0.90$ AND derived from S_{prov}	Cryptographic proof confirms human/camera origin; no AI edits detected.
Suspicious	$m(\text{Synthetic}) > \text{Thresh}$, driven primarily by S_{dl}	Metadata/watermarks stripped, but neural analysis indicates deepfake artifacts.
Unknown	$m(\Omega) > \text{Threshold}$	No provenance found, and DL forensic signals are ambiguous or contradictory.

System Architecture and Implementation Framework

To transition this theoretical framework into a functional, deployable tool capable of processing media at scale, a robust software and hardware architecture is required. The system design strictly adheres to the requirements necessary to support deep learning inference, cryptographic validation, and rapid user interaction⁷.

The underlying hardware infrastructure requires significant parallel processing capabilities to execute the 3D CNN and Vision Transformer pipelines with acceptable latency. The minimum viable deployment architecture necessitates a multi-core processor (Intel i5 or AMD Ryzen 5 tier or higher) paired with a minimum of 16 GB of system RAM. Crucially, the system relies on GPU acceleration for tensor operations, requiring an NVIDIA GPU with at least 6 GB of VRAM to comfortably load the parameter-heavy XceptionNet and DINOv3 foundation models in sequence⁷. High-speed local storage (minimum 512 GB NVMe recommended) is required to manage the caching of uncompressed video frames during the spatiotemporal analysis phase⁷. The software stack is built entirely on Python 3.x, utilizing PyTorch as the primary deep learning framework for model execution and inference. The media preprocessing pipeline—responsible for image decoding, dynamic frame extraction, resizing, and normalization—is driven by OpenCV and the Pillow library⁷. Numerical manipulation and the execution of the Dempster-Shafer matrix calculations are handled by NumPy and SciPy, while Pandas manages the structured logging of the forensic outputs⁷.

The system architecture is bifurcated into a decoupled front-end interface and a high-performance back-end processing engine. The front-end, designed for rapid research prototyping and user interaction, is built using Streamlit. This interface allows users to seamlessly upload image or video assets and view the resulting authentication status, displaying the individual confidence scores for provenance, watermarking, and forensics alongside the fused DST output⁷. The back-end evidence fusion engine operates as an asynchronous microservice via FastAPI. This decoupling allows the computationally expensive deep learning models to process media sequentially in the background, utilizing Git and GitHub for continuous version control and integration of updated forensic weights⁷. The FastAPI backend orchestrates the three distinct analysis phases, queries external C2PA Manifest Stores for soft binding resolution, and returns the final JSON payload containing the unified authentication classification to the Streamlit frontend.

Conclusion

The authentication of digital media in the era of advanced generative AI cannot rely on a singular technological mechanism. The pursuit of a "silver bullet" deepfake detector is structurally flawed, as generative algorithms will continually evolve to minimize the statistical anomalies detected by static neural networks. Cryptographic provenance architectures, specifically the C2PA standard, provide the necessary transparency, chain-of-custody tracking, and regulatory compliance required by modern legislation such as the EU AI Act. However, these systems fail instantly upon the routine metadata stripping executed by nearly all

consumer distribution platforms. Invisible digital watermarking bridges this gap by embedding resilient soft bindings directly into the perceptual payload, yet it remains fundamentally vulnerable to sophisticated diffusion-based adversarial purification and optimization attacks. Deep learning forensic detectors offer a vital, uncooperative safety net, but are historically prone to cross-generator generalization failures and high false-positive rates when deployed in isolation.

The hybrid framework detailed in this report resolves these isolated vulnerabilities through structural redundancy and probabilistic evidence fusion. By continuously and simultaneously extracting signals across the cryptographic, steganographic, and intrinsic feature domains, and subsequently synthesizing those signals via Dempster-Shafer Theory, the system effectively neutralizes single-vector evasion attacks. The mathematical fusion ensures that if a malicious actor strips the manifest, the watermark recovers the provenance; if the watermark is adversarially erased, the temporal artifacts introduced by the erasure process trigger the spatiotemporal forensic detector. This multi-layered, mathematically rigorous approach provides a highly robust, scalable, and operationally viable defense against the escalating threat of weaponized synthetic media. Future research in this domain must prioritize optimizing the computational overhead of the 3D CNN and Vision Transformer pipelines to facilitate real-time edge deployment, ensuring that this defense-in-depth architecture can operate seamlessly at the point of media consumption across the global digital ecosystem.

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